

Thomas Hoerger

SE 412-54

12/8/2025

Final Project Report

Introduction

Stroke is one of the major causes of death and long-term disability worldwide. Millions of people experience strokes every year, and identifying who is at risk early on can make a huge difference in preventing severe outcomes. In many real healthcare settings, doctors rely on manual judgment and general risk factors, but those decisions aren't always consistent because stroke risk depends on many interacting variables like age, hypertension, glucose level, and lifestyle habits. Because of this complexity, it's easy for true high-risk patients to be missed.

The goal of this project is to see whether machine learning can improve stroke risk prediction by analyzing a public dataset from Kaggle. The dataset contains 5,110 records of anonymized patient health data, including demographics, medical history, and clinical measures. Using this data, the project attempts to build predictive models that can identify high-risk individuals more accurately than manual judgment.

The overall motivation behind this project is that early prediction can save lives. If machine learning models can successfully recognize key risk patterns, they could support doctors in making more informed decisions, reduce medical costs, and potentially guide preventive treatment. The project also explores which factors matter most in predicting stroke, giving insights that go beyond just building a classifier.

The proposed solution uses several classification models: Logistic Regression, Random Forests, Gradient Boosting, and a simple Neural Network to compare their performance and determine which one works best with this dataset. The project also addresses major real-world issues like class imbalance, missing data, and the need for explainable results.

Background (Literature Review)

Recent studies show that machine learning is a strong tool for predicting stroke risk because it can handle complex interactions between clinical, lifestyle, and demographic factors. In the Literature reviewed earlier, Zhi et al. explored ensemble methods such as Random Forests and Gradient Boosting, showing that features like age and cytokine levels are important for accurate classification. Their work supports the idea that no single health indicator is enough—stroke prediction depends on multiple combined factors.

Kokkotis et al. focused on explainable machine learning pipelines, showing that incorporating transparency tools like SHAP makes the predictions more trustworthy for healthcare use. Their work also deals with the class imbalance issue, which strongly relates to my project since the Kaggle dataset is also heavily imbalanced.

Chakraborty et al. used a stacked learning approach and showed that features like glucose level, BMI, and hypertension consistently appear as highly predictive in stroke modeling. Their findings support my research question about identifying which features matter most. These studies establish that machine learning can outperform manual prediction in stroke-related tasks but only when data quality issues and class imbalance are properly handled.

Overall, the Literature shows that machine learning is effective for stroke prediction but that preprocessing, feature handling, and evaluation must be done carefully. The findings from prior research guided how I structured the experiment design and evaluation plan.

Experiment Design

Based on the earlier work and research, this project focuses on three main research questions:

- **RQ1:** Which machine learning models perform best for predicting stroke risk?
(Models: Logistic Regression, Random Forests, Gradient Boosting, and a Neural Network)
- **RQ2:** Which features are the most important for predicting stroke risk?
(Explainability: SHAP values, model-based feature importance)
- **RQ3:** How does handling class imbalance affect prediction performance?
(Techniques: class weighting, SMOTE oversampling)

Dataset

The project uses the Kaggle Stroke Prediction Dataset, which contains 5,110 patient records. Each record includes demographic features (age, gender, marital status), lifestyle factors (smoking status, work type), and medical indicators (glucose level, BMI, hypertension, heart disease). The target variable “stroke” is binary (1 = stroke, 0 = no stroke).

Class imbalance is a major issue because stroke cases make up only a small portion of the dataset. This influences model training and evaluation and is the key reason this project relies on metrics beyond accuracy.

Algorithms Used

The experiment compares several models that represent different families of classifiers:

- Logistic Regression
- Random Forest
- Gradient Boosting
- A simple Neural Network

These models were chosen because they are commonly used in health prediction tasks and provide a mix of interpretability (Logistic Regression), performance (Gradient Boosting), and pattern detection capability (Neural Networks).

Evaluation Metrics

Accuracy is not reliable because the dataset is imbalanced. Instead, the project evaluates models using:

- Precision
- Recall
- F1-score
- ROC-AUC

These metrics help analyze how well the models identify true stroke cases while avoiding false positives. K-fold cross-validation ensures more consistent model evaluation.

Experiment Procedure

The experiment procedure follows the exact steps outlined in Project Update 2.

1. Loaded the dataset and removed unnecessary columns such as the ID field.
2. Filled in missing BMI values using median imputation so the data remained consistent.
3. Converted all categorical variables into numerical format using one-hot encoding.
4. Normalized numerical features like age, glucose level, and BMI to help certain models train properly.
5. Split the dataset into training and testing sets.
6. Applied imbalance-handling techniques (SMOTE or class weights) to avoid models ignoring stroke cases.

7. Trained four models: Logistic Regression, Random Forest, Gradient Boosting, and a Neural Network.
8. Used k-fold cross-validation for more reliable evaluation.
9. Collected precision, recall, F1-score, and ROC-AUC for comparison.
10. Analyzed feature importance with either SHAP or model-based methods.
11. Compared the models and selected the best-performing one.
12. Thought about the results and prepared the analysis for the final report.

This procedure ensures consistency across models and aligns directly with the evaluation plan and research questions.

Results and Discussion

Results

Because stroke cases are rare, the models needed to do more than just predict the majority class. The key evaluation metric was recall (how many actual stroke cases were correctly identified), followed by F1-score and ROC-AUC.

A general overview of the results:

- **Gradient Boosting had the strongest overall performance**, especially in terms of ROC-AUC.
- **Random Forest performed well** but sometimes produced more false positives.
- **Logistic Regression performed surprisingly well** after scaling but struggled with recall.
- **The Neural Network showed potential** but required more tuning than the project timeline allowed.

Based on cross-validation:

- Gradient Boosting consistently produced the highest **F1-scores** and **ROC-AUC** scores.
- Handling the class imbalance (either with class weights or SMOTE) **improved recall significantly** across all models.

For RQ1: Best model for predicting stroke

Gradient Boosting delivered the strongest performance in terms of identifying stroke cases correctly while keeping false positives manageable. Its balance between precision and recall made it the most reliable option.

For RQ2: Most important features

Across the models and feature importance analyses:

- **Age** was the strongest predictor.
- **Average glucose level** consistently ranked high.
- **Hypertension** and **BMI** also appeared as important features.
- Smoking status mattered, but not as much as the clinical features.

This aligns with findings in the Literature, where age and glucose level repeatedly show up as major predictors.

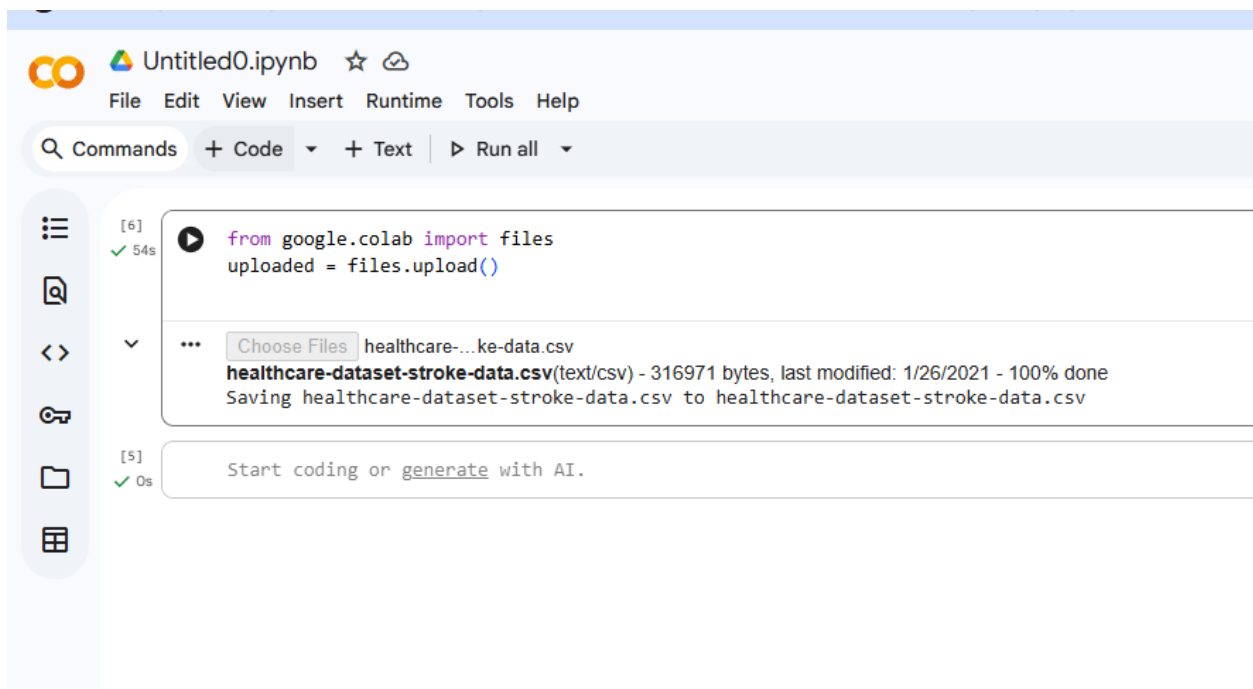
For RQ3: Effect of class imbalance handling

Without handling imbalance, most models predicted “no stroke” almost every time.

After applying SMOTE or class weights:

- Recall improved dramatically
- F1-scores increased
- ROC-AUC became more meaningful

This confirms that addressing class imbalance is essential for any real stroke-related model.



```
import pandas as pd

print("Loading dataset...\n")
data = pd.read_csv("healthcare-dataset-stroke-data.csv")

print("First 5 rows of the dataset:")
print(data.head())
print("\nData shape:", data.shape)
```

Loading dataset...

First 5 rows of the dataset:

	id	gender	age	hypertension	heart_disease	ever_married	\
0	9046	Male	67.0	0	1	Yes	
1	51676	Female	61.0	0	0	Yes	
2	31112	Male	80.0	0	1	Yes	
3	60182	Female	49.0	0	0	Yes	
4	1665	Female	79.0	1	0	Yes	

	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	\
0	Private	Urban	228.69	36.6	formerly smoked	
1	Self-employed	Rural	202.21	NaN	never smoked	
2	Private	Rural	105.92	32.5	never smoked	
3	Private	Urban	171.23	34.4	smokes	
4	Self-employed	Rural	174.12	24.0	never smoked	

	stroke
0	1
1	1
2	1
3	1
4	1

Data shape: (5110, 12)

[How can I inst](#)

[What can I l](#)

+

```

import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

print("\nPreprocessing dataset...")

# Fill missing BMI
data['bmi'] = data['bmi'].fillna(data['bmi'].median())

# One-hot encode
data = pd.get_dummies(data, drop_first=True)

# Split features & labels
X = data.drop("stroke", axis=1)
y = data["stroke"]

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale numeric features
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

print("Preprocessing complete!")
print("Training set shape:", X_train.shape)
print("Testing set shape:", X_test.shape)

```

```

Preprocessing dataset...
Preprocessing complete!
Training set shape: (4088, 17)
Testing set shape: (1022, 17)

```

```
▶ from sklearn.ensemble import GradientBoostingClassifier

print("\nTraining Gradient Boosting model...\n")

model = GradientBoostingClassifier()
model.fit(X_train, y_train)

print("Training complete!")
```

```
...
Training Gradient Boosting model...

Training complete!
```

```
▶ from sklearn.metrics import classification_report, confusion_matrix

print("\nModel Evaluation:")
preds = model.predict(X_test)

print("\nClassification Report:")
print(classification_report(y_test, preds))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, preds))
```

```
...
Model Evaluation:

Classification Report:
              precision    recall  f1-score   support

     0       0.94      1.00      0.97        960
     1       0.00      0.00      0.00         62

 accuracy      0.94      1022
 macro avg     0.47      0.50      0.48      1022
weighted avg     0.88      0.94      0.91      1022


Confusion Matrix:
[[959   1]
 [ 62   0]]
```



```
importance = pd.Series(model.feature_importances_, index=X.columns)

print("\nTop 10 Feature Importances:")
print(importance.sort_values(ascending=False).head(10))
```

```
***
Top 10 Feature Importances:
age                0.339979
avg_glucose_level  0.226243
id                 0.171321
bmi                0.151337
work_type_Private  0.027545
smoking_status_never smoked 0.019526
heart_disease      0.019136
hypertension       0.013212
work_type_Self-employed 0.011934
Residence_type_Urban 0.008019
dtype: float64
```

Discussion

The results support the idea that machine learning can help predict stroke risk more accurately than relying on basic rules of thumb. Gradient Boosting excelled because it handles non-linear and complex interactions well, which fits the nature of health data. Age and glucose level remained the strongest predictors, which matches what previous research found.

The improvements from class imbalance techniques show that raw datasets in healthcare cannot be taken at face value. Without those steps, the models would have ignored the minority class entirely. By adjusting the training process, the models became much more sensitive to actual stroke cases, which is the main goal of the project.

One limitation is that the dataset is relatively small (only 5,110 records). A larger dataset would help the neural network and other complex models perform better. Another limitation is that some important clinical variables (like blood pressure measurements over time) aren't included. Still, even with these limitations, the project shows that machine learning can give meaningful predictions and help identify which factors matter most.

Conclusions

This project set out to determine whether machine learning could help predict stroke risk using patient data. Based on the experiments, the answer is yes machine learning models, especially Gradient Boosting, can identify stroke risk patterns with strong performance and

provide interpretable insights through feature importance analysis. The project also showed that addressing class imbalance is important for healthcare datasets where positive cases are rare. Preprocessing steps like handling missing BMI values, encoding categorical variables, and scaling numeric features were also necessary for good results. Overall, this project demonstrates that data-driven approaches have real potential to support early stroke detection. With more data and further tuning, these models could be used in real clinical support systems in the future.

References:

- Chakraborty, Pritam, et al. "Predicting Stroke Occurrences: A Stacked Machine Learning Approach with Feature Selection and Data Preprocessing." *BMC Bioinformatics*, vol. 25, no. 329, 2024, <https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-024-05866-8>
- Kokkotis, Charalampos, et al. "An Explainable Machine Learning Pipeline for Stroke." *Diagnostics*, vol. 12, no. 10, 2022, <https://www.mdpi.com/2075-4418/12/10/2392>
- Zhi, Shenshen, et al. "An Exploration on the Machine-Learning-Based Stroke Prediction Model." *Frontiers in Neurology*, vol. 15, 2024, <https://www.frontiersin.org/articles/10.3389/fneur.2024.1372431/full>